PREMIUM

Full-Length Mock Test

Mock 01

Biotechnology

GATE BT

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | Moderate 33% | Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Bioprocess Technology — Kinetics]

Enzyme classification (EC) has how many classes?

A) 4
B) 6
C) 8
D) 10

Q.2 [1 Mark] [EASY] [Biochemistry — Enzymes]

PCR was developed by:

A) Watson & Crick
B) Kary Mullis
C) Frederick Sanger
D) Linus Pauling

Q.3 [1 Mark] [EASY] [Microbiology — Culture Media]

Bacterial cell wall is made of:

A) Cellulose
B) Peptidoglycan
C) Chitin
D) Starch

Q.4 [1 Mark] [EASY] [Molecular Biology — Genetic Engineering]

Plasmid is:

A) Chromosomal DNA
B) Extrachromosomal circular DNA
C) mRNA
D) Ribosome

Q.5 [1 Mark] [EASY] [Genetics — Mutation]

Enzyme classification (EC) has how many classes?

A) 4
B) 6
C) 8
D) 10

Q.6 [1 Mark] [EASY] [Bioprocess Technology — Downstream Processing]

PCR was developed by:

A) Watson & Crick
B) Kary Mullis
C) Frederick Sanger
D) Linus Pauling

Q.7 [1 Mark] [EASY] [Biochemistry — Metabolism]

Bacterial cell wall is made of:

A) Cellulose

B) Peptidoglycan

C) Chitin

D) Starch

Q.8 [1 Mark] [EASY] [Microbiology — Viruses]

Plasmid is:

A) Chromosomal DNA
B) Extrachromosomal circular DNA
C) mRNA
D) Ribosome

Q.9 [1 Mark] [EASY] [Molecular Biology — Translation]

Enzyme classification (EC) has how many classes?

A) 4
B) 6
C) 8
D) 10

Q.10 [1 Mark] [EASY] [Genetics — Mendelian Genetics]

PCR was developed by:

A) Watson & Crick
B) Kary Mullis
C) Frederick Sanger
D) Linus Pauling

Q.11 [1 Mark] [EASY] [Bioprocess Technology — Bioreactors]

Bacterial cell wall is made of:

A) Cellulose
B) Peptidoglycan
C) Chitin
D) Starch

Q.12 [1 Mark] [EASY] [Biochemistry — Bioenergetics]

Plasmid is:

A) Chromosomal DNA
B) Extrachromosomal circular DNA
C) mRNA
D) Ribosome

Q.13 [2 Marks] [MODERATE] [Microbiology — Viruses]

Enzyme classification (EC) has how many classes?

A) 4
B) 6
C) 8
D) 10

Q.14 [2 Marks] [MODERATE] [Molecular Biology — Transcription]

PCR was developed by:

A) Watson & Crick

B) Kary Mullis

C) Frederick Sanger

D) Linus Pauling

Q.15 [2 Marks] [MODERATE] [Genetics — Mutation]

Bacterial cell wall is made of:

A) Cellulose
B) Peptidoglycan
C) Chitin
D) Starch

Q.16 [2 Marks] [MODERATE] [Bioprocess Technology — Oxygen Transfer]

Plasmid is:

A) Chromosomal DNA
B) Extrachromosomal circular DNA
C) mRNA
D) Ribosome

Q.17 [2 Marks] [MODERATE] [Biochemistry — Metabolism]

Enzyme classification (EC) has how many classes?

A) 4
B) 6
C) 8
D) 10

Q.18 [2 Marks] [MODERATE] [Microbiology — Immunology]

PCR was developed by:

A) Watson & Crick
B) Kary Mullis
C) Frederick Sanger
D) Linus Pauling

Q.19 [2 Marks] [MODERATE] [Molecular Biology — Genetic Engineering]

Bacterial cell wall is made of:

A) Cellulose
B) Peptidoglycan
C) Chitin
D) Starch

Q.20 [2 Marks] [MODERATE] [Genetics — Genomics]

Plasmid is:

A) Chromosomal DNA
B) Extrachromosomal circular DNA
C) mRNA
D) Ribosome

Q.21 [2 Marks] [MODERATE] [Bioprocess Technology — Scale-up]

Enzyme classification (EC) has how many classes?

A) 4

B) 6

C) 8

D) 10

Q.22 [2 Marks] [MODERATE] [Biochemistry — Carbohydrates]

PCR was developed by:

A) Watson & Crick
B) Kary Mullis
C) Frederick Sanger
D) Linus Pauling

Q.23 [2 Marks] [MODERATE] [Microbiology — Bacteria]

Bacterial cell wall is made of:

A) Cellulose
B) Peptidoglycan
C) Chitin
D) Starch

Q.24 [2 Marks] [MODERATE] [Molecular Biology — PCR]

Plasmid is:

A) Chromosomal DNA
B) Extrachromosomal circular DNA
C) mRNA
D) Ribosome

Q.25 [2 Marks] [MODERATE] [Genetics — Genomics]

Enzyme classification (EC) has how many classes?

A) 4
B) 6
C) 8
D) 10

Q.26 [2 Marks] [MODERATE] [Bioprocess Technology — Downstream Processing]

PCR was developed by:

A) Watson & Crick
B) Kary Mullis
C) Frederick Sanger
D) Linus Pauling

Q.27 [2 Marks] [MODERATE] [Biochemistry — Enzymes]

Bacterial cell wall is made of:

A) Cellulose
B) Peptidoglycan
C) Chitin
D) Starch

Q.28 [2 Marks] [MODERATE] [Microbiology — Sterilization]

Plasmid is:

A) Chromosomal DNA

B) Extrachromosomal circular DNA

C) mRNA

D) Ribosome

Q.29 [2 Marks] [MODERATE] [Molecular Biology — DNA Replication]

Enzyme classification (EC) has how many classes?

A) 4
B) 6
C) 8
D) 10

Q.30 [2 Marks] [MODERATE] [Genetics — Recombination]

PCR was developed by:

A) Watson & Crick
B) Kary Mullis
C) Frederick Sanger
D) Linus Pauling

Q.31 [2 Marks] [MODERATE] [Bioprocess Technology — Kinetics]

Bacterial cell wall is made of:

A) Cellulose
B) Peptidoglycan
C) Chitin
D) Starch

Q.32 [2 Marks] [MODERATE] [Biochemistry — Enzymes]

Plasmid is:

A) Chromosomal DNA
B) Extrachromosomal circular DNA
C) mRNA
D) Ribosome

Q.33 [2 Marks] [HARD] [Microbiology — Viruses]

Enzyme classification (EC) has how many classes?

A) 4
B) 6
C) 8
D) 10

Q.34 [2 Marks] [HARD] [Molecular Biology — Translation]

PCR was developed by:

A) Watson & Crick
B) Kary Mullis
C) Frederick Sanger
D) Linus Pauling

Q.35 [2 Marks] [HARD] [Genetics — Recombination]

Bacterial cell wall is made of:

A) Cellulose

B) Peptidoglycan

C) Chitin

D) Starch

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Bioprocess Technology — Kinetics]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.37 [2 Marks] [HARD] [Biochemistry — Proteins]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.38 [2 Marks] [HARD] [Microbiology — Culture Media]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.39 [2 Marks] [HARD] [Molecular Biology — Translation]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.40 [2 Marks] [HARD] [Genetics — Genomics]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Bioprocess Technology — Oxygen Transfer]

Enzyme Km = 0.5mM, Vmax = 100;ÄÒöÖ–âà At [S]=1mM, rate "H ?

Answer: ______________________

Q.52 [2 Marks] [HARD] [Biochemistry — Metabolism]

E. coli doubling time ~20min. In 2hrs, cells increase by factor = ?

Answer: ______________________

Q.53 [2 Marks] [HARD] [Microbiology — Culture Media]

Enzyme Km = 0.5mM, Vmax = 100;ÄÒöÖ–âà At [S]=1mM, rate "H ?

Answer: ______________________

Q.54 [2 Marks] [HARD] [Molecular Biology — Translation]

E. coli doubling time ~20min. In 2hrs, cells increase by factor = ?

Answer: ______________________

Q.55 [2 Marks] [HARD] [Genetics — Mutation]

Enzyme Km = 0.5mM, Vmax = 100;ÄÒöÖ–âà At [S]=1mM, rate "H ?

Answer: ______________________

Q.56 [2 Marks] [HARD] [Bioprocess Technology — Downstream Processing]

E. coli doubling time ~20min. In 2hrs, cells increase by factor = ?

Answer: ______________________

Q.57 [2 Marks] [HARD] [Biochemistry — Bioenergetics]

Enzyme Km = 0.5mM, Vmax = 100;ÄÒöÖ–âà At [S]=1mM, rate "H ?

Answer: ______________________

Q.58 [2 Marks] [HARD] [Microbiology — Culture Media]

E. coli doubling time ~20min. In 2hrs, cells increase by factor = ?

Answer: ____________________

Q.59 [2 Marks] [HARD] [Molecular Biology — Translation]

Enzyme Km = 0.5mM, Vmax = 100;ÄÒöÖ–âà At [S]=1mM, rate "H ?

Answer: ____________________________

Q.60 [2 Marks] [HARD] [Genetics — Mutation]

E. coli doubling time ~20min. In 2hrs, cells increase by factor = ?

Answer: ____________________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	B	Q.2:	B	Q.3:	B
Q.4:	B	Q.5:	B	Q.6:	B
Q.7:	B	Q.8:	B	Q.9:	B
Q.10:	B	Q.11:	B	Q.12:	B
Q.13:	B	Q.14:	B	Q.15:	B
Q.16:	B	Q.17:	B	Q.18:	B
Q.19:	B	Q.20:	B	Q.21:	B
Q.22:	B	Q.23:	B	Q.24:	B
Q.25:	B	Q.26:	B	Q.27:	B
Q.28:	B	Q.29:	B	Q.30:	B
Q.31:	B	Q.32:	B	Q.33:	B
Q.34:	B	Q.35:	B	Q.36:	AC
Q.37:	AC	Q.38:	AC	Q.39:	AC
Q.40:	AC	Q.41:	AC	Q.42:	AC
Q.43:	AC	Q.44:	AC	Q.45:	AC
Q.46:	AC	Q.47:	AC	Q.48:	AC
Q.49:	AC	Q.50:	AC	Q.51:	67
Q.52:	64	Q.53:	67	Q.54:	64
Q.55:	67	Q.56:	64	Q.57:	67
Q.58:	64	Q.59:	67	Q.60:	64

Detailed Solutions

Question 1 [Bioprocess Technology — Kinetics]

Enzyme classification (EC) has how many classes?

Answer: (B)

Six enzyme classes: oxidoreductases, transferases, hydrolases, lyases, isomerases, ligases.

Question 2 [Biochemistry — Enzymes]

PCR was developed by:

Answer: (B)

Kary Mullis invented PCR in 1983, Nobel Prize 1993.

Question 3 [Microbiology — Culture Media]

Bacterial cell wall is made of:

Answer: (B)

Bacterial cell wall is peptidoglycan (murein).

Question 4 [Molecular Biology — Genetic Engineering]

Plasmid is:

Answer: (B)

Plasmids are circular, self-replicating extrachromosomal DNA molecules.

Question 5 [Genetics — Mutation]

Enzyme classification (EC) has how many classes?

Answer: (B)

Six enzyme classes: oxidoreductases, transferases, hydrolases, lyases, isomerases, ligases.

Question 6 [Bioprocess Technology — Downstream Processing]

PCR was developed by:

Answer: (B)

Kary Mullis invented PCR in 1983, Nobel Prize 1993.

Question 7 [Biochemistry — Metabolism]

Bacterial cell wall is made of:

Answer: (B)

Bacterial cell wall is peptidoglycan (murein).

Question 8 [Microbiology — Viruses]

Plasmid is:

Answer: (B)

Plasmids are circular, self-replicating extrachromosomal DNA molecules.

Question 9 [Molecular Biology — Translation]

Enzyme classification (EC) has how many classes?

Answer:

(B)

Six enzyme classes: oxidoreductases, transferases, hydrolases, lyases, isomerases, ligases.

Question 10
[Genetics — Mendelian Genetics]

PCR was developed by:

Answer: (B)

Kary Mullis invented PCR in 1983, Nobel Prize 1993.

Question 11
[Bioprocess Technology — Bioreactors]

Bacterial cell wall is made of:

Answer: (B)

Bacterial cell wall is peptidoglycan (murein).

Question 12
[Biochemistry — Bioenergetics]

Plasmid is:

Answer: (B)

Plasmids are circular, self-replicating extrachromosomal DNA molecules.

Question 13
[Microbiology — Viruses]

Enzyme classification (EC) has how many classes?

Answer: (B)

Six enzyme classes: oxidoreductases, transferases, hydrolases, lyases, isomerases, ligases.

Question 14
[Molecular Biology — Transcription]

PCR was developed by:

Answer: (B)

Kary Mullis invented PCR in 1983, Nobel Prize 1993.

Question 15
[Genetics — Mutation]

Bacterial cell wall is made of:

Answer: (B)

Bacterial cell wall is peptidoglycan (murein).

Question 16
[Bioprocess Technology — Oxygen Transfer]

Plasmid is:

Answer: (B)

Plasmids are circular, self-replicating extrachromosomal DNA molecules.

Question 17
[Biochemistry — Metabolism]

Enzyme classification (EC) has how many classes?

Answer: (B)

Six enzyme classes: oxidoreductases, transferases, hydrolases, lyases, isomerases, ligases.

Question 18
[Microbiology — Immunology]

PCR was developed by:

Answer:

(B)

Kary Mullis invented PCR in 1983, Nobel Prize 1993.

Question 19 *[Molecular Biology — Genetic Engineering]*

Bacterial cell wall is made of:

Answer: (B)

Bacterial cell wall is peptidoglycan (murein).

Question 20 *[Genetics — Genomics]*

Plasmid is:

Answer: (B)

Plasmids are circular, self-replicating extrachromosomal DNA molecules.

Question 21 *[Bioprocess Technology — Scale-up]*

Enzyme classification (EC) has how many classes?

Answer: (B)

Six enzyme classes: oxidoreductases, transferases, hydrolases, lyases, isomerases, ligases.

Question 22 *[Biochemistry — Carbohydrates]*

PCR was developed by:

Answer: (B)

Kary Mullis invented PCR in 1983, Nobel Prize 1993.

Question 23 *[Microbiology — Bacteria]*

Bacterial cell wall is made of:

Answer: (B)

Bacterial cell wall is peptidoglycan (murein).

Question 24 *[Molecular Biology — PCR]*

Plasmid is:

Answer: (B)

Plasmids are circular, self-replicating extrachromosomal DNA molecules.

Question 25 *[Genetics — Genomics]*

Enzyme classification (EC) has how many classes?

Answer: (B)

Six enzyme classes: oxidoreductases, transferases, hydrolases, lyases, isomerases, ligases.

Question 26 *[Bioprocess Technology — Downstream Processing]*

PCR was developed by:

Answer: (B)

Kary Mullis invented PCR in 1983, Nobel Prize 1993.

Question 27 *[Biochemistry — Enzymes]*

Bacterial cell wall is made of:

Answer:

(B)

Bacterial cell wall is peptidoglycan (murein).

Question 28 *[Microbiology — Sterilization]*

Plasmid is:

Answer: (B)

Plasmids are circular, self-replicating extrachromosomal DNA molecules.

Question 29 *[Molecular Biology — DNA Replication]*

Enzyme classification (EC) has how many classes?

Answer: (B)

Six enzyme classes: oxidoreductases, transferases, hydrolases, lyases, isomerases, ligases.

Question 30 *[Genetics — Recombination]*

PCR was developed by:

Answer: (B)

Kary Mullis invented PCR in 1983, Nobel Prize 1993.

Question 31 *[Bioprocess Technology — Kinetics]*

Bacterial cell wall is made of:

Answer: (B)

Bacterial cell wall is peptidoglycan (murein).

Question 32 *[Biochemistry — Enzymes]*

Plasmid is:

Answer: (B)

Plasmids are circular, self-replicating extrachromosomal DNA molecules.

Question 33 *[Microbiology — Viruses]*

Enzyme classification (EC) has how many classes?

Answer: (B)

Six enzyme classes: oxidoreductases, transferases, hydrolases, lyases, isomerases, ligases.

Question 34 *[Molecular Biology — Translation]*

PCR was developed by:

Answer: (B)

Kary Mullis invented PCR in 1983, Nobel Prize 1993.

Question 35 *[Genetics — Recombination]*

Bacterial cell wall is made of:

Answer: (B)

Bacterial cell wall is peptidoglycan (murein).

Question 36 *[Bioprocess Technology — Kinetics]*

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 37 *[Biochemistry — Proteins]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 38 *[Microbiology — Culture Media]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 39 *[Molecular Biology — Translation]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 40 *[Genetics — Genomics]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 41 *[Bioprocess Technology — Downstream Processing]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 42 *[Biochemistry — Bioenergetics]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 43 *[Microbiology — Sterilization]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 44 *[Molecular Biology — Translation]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 45 *[Genetics — Mutation]*

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 46 *[Bioprocess Technology — Scale-up]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 47 *[Biochemistry — Bioenergetics]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 48 *[Microbiology — Sterilization]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 49 *[Molecular Biology — PCR]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 50 *[Genetics — Genomics]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 51 *[Bioprocess Technology — Oxygen Transfer]*

Enzyme Km = 0.5mM, Vmax = 100;ÄÒöÖ–âà At [S]=1mM, rate "H ?

Answer: (67)

v = Vmax[S]/(Km+[S]) = 100×1/(0.5+1) = 66.67 ;ÄÒöÖ–âà

Question 52 *[Biochemistry — Metabolism]*

E. coli doubling time ~20min. In 2hrs, cells increase by factor = ?

Answer: (64)

2hr = 120min. Generations = 120/20 = 6. Factor = 2^6 = 64.

Question 53 *[Microbiology — Culture Media]*

Enzyme Km = 0.5mM, Vmax = 100;ÄÒöÖ–âà At [S]=1mM, rate "H ?

Answer: (67)

v = Vmax[S]/(Km+[S]) = 100×1/(0.5+1) = 66.67 ;ÄÒöÖ–âà

Question 54 *[Molecular Biology — Translation]*

E. coli doubling time ~20min. In 2hrs, cells increase by factor = ?

Answer:

2hr = 120min. Generations = 120/20 = 6. Factor = 2^6 = 64.

Question 55 *[Genetics — Mutation]*

Enzyme Km = 0.5mM, Vmax = 100;ÄÒöÖ–âà At [S]=1mM, rate "H ?

Answer: (67)

v = Vmax[S]/(Km+[S]) = 100×1/(0.5+1) = 66.67 ;ÄÒöÖ–âà

Question 56 *[Bioprocess Technology — Downstream Processing]*

E. coli doubling time ~20min. In 2hrs, cells increase by factor = ?

Answer: (64)

2hr = 120min. Generations = 120/20 = 6. Factor = 2^6 = 64.

Question 57 *[Biochemistry — Bioenergetics]*

Enzyme Km = 0.5mM, Vmax = 100;ÄÒöÖ–âà At [S]=1mM, rate "H ?

Answer: (67)

v = Vmax[S]/(Km+[S]) = 100×1/(0.5+1) = 66.67 ;ÄÒöÖ–âà

Question 58 *[Microbiology — Culture Media]*

E. coli doubling time ~20min. In 2hrs, cells increase by factor = ?

Answer: (64)

2hr = 120min. Generations = 120/20 = 6. Factor = 2^6 = 64.

Question 59 *[Molecular Biology — Translation]*

Enzyme Km = 0.5mM, Vmax = 100;ÄÒöÖ–âà At [S]=1mM, rate "H ?

Answer: (67)

v = Vmax[S]/(Km+[S]) = 100×1/(0.5+1) = 66.67 ;ÄÒöÖ–âà

Question 60 *[Genetics — Mutation]*

E. coli doubling time ~20min. In 2hrs, cells increase by factor = ?

Answer: (64)

2hr = 120min. Generations = 120/20 = 6. Factor = 2^6 = 64.
